When Best Science Becomes Opaque: Hybrid Science and the Non-Naturalistic Metaphysics of Scientific Practice

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Abstract

Naturalistic metaphysics of science often proceeds from a deferential assumption: metaphysical inquiry should take its ontological, epistemic, and methodological orientation from our best science. This paper argues that AI-mediated science makes that assumption increasingly problematic. The difficulty is not that science has ceased to be authoritative, but that "best science" is no longer a transparent object of deference. Digital systems increasingly participate not only in calculation or instrumentation, but in classification, pattern detection, model construction, anomaly recognition, prediction, and the stabilization of problem-spaces. Scientific rationality is therefore becoming distributed across human and machine components in ways that are often opaque to the human communities that rely on them.

This emerging formation can best be described as a mode of inquiry in which the production, stabilization, and evaluation of scientific objects, facts, and explanations are distributed across human agents, machine systems, datasets, models, instruments, and institutions. It is not simply science plus AI, but a transitional regime in which the constraint-structure of scientific rationality is redistributed across human and machine systems. To clarify this transitional status, the paper uses two examples: string theory as a case in which scientific authority becomes detached from direct empirical confirmation, and AI-mediated science as a case in which scientific authority becomes entangled with procedural opacity. I call this formation **hybrid science**.

Drawing on Thomas Kuhn's account of paradigms and disciplinary matrices, Ludwik Fleck's theory of thought collectives, and Karl Mannheim's perspectival sociology of knowledge, I argue that scientific authority depends on historically generated, collectively stabilized, perspectivally organized, and professionally transmitted constraints. I call this structure **constrained rational normativity**: the form of rational binding through which science remains authoritative without resting on timeless method, neutral observation, or algorithmic rules. By connecting string theory's status as a paradigm-in-transition with AI-mediated opacity, the paper argues that contemporary science increasingly operates through transitional regimes of constrained rational normativity.

AI-mediated opacity therefore reveals a limit in deferential naturalism. If metaphysics is told to defer to the best science, it must first ask what science has become when its world-disclosing practices are historically constituted, technologically mediated, and partially opaque. A non-naturalistic metaphysics of science is not anti-scientific; it is a critical inquiry into the conditions under which scientific authority, objectivity, and ontology are generated, stabilized, revised, and made accountable. At the end of this trajectory, science is best understood as a historically evolving system of constrained world-disclosure. In the age of AI, that system becomes hybrid.